

Introduction

The system aims to restore ridge flow patterns, enhance image clarity, and isolate ridge structures so that the fingerprint can be analysed more accurately. This matches the assignment scope, which requires the fingerprint topic to focus on ridge restoration, clarity improvement, and ridge structure isolation for feature analysis.

Problem Statement

Low-quality fingerprint images are difficult to analyse because the **ridge and valley patterns** may be affected by noise, blur, low contrast, incomplete impressions, and broken ridge structures. These problems can reduce fingerprint clarity and make important features such as **ridge endings**, bifurcations, and skeletonised ridge patterns harder to extract accurately.

Although existing fingerprint **enhancement** techniques such as Adaptive Histogram Equalisation, Gabor filtering, Modified Gabor filtering, and morphological processing can improve ridge visibility, their performance may still be affected by **severe noise and unclear ridge flow**. Some techniques may enhance contrast but **also amplify noise**, while others may improve ridge structure but fail to fully restore broken or degraded ridge patterns.

Therefore, there is a need to compare suitable image processing techniques for fingerprint enhancement and identify which combination produces clearer ridge structures. This project aims to develop a fingerprint enhancement pipeline that applies preprocessing, ridge enhancement, segmentation, morphological operations, thinning, and feature extraction. The enhancement result will be evaluated **using PSNR, MSE, SSIM**, ridge clarity, and feature extraction quality.

Objectives

1. **To enhance low-quality fingerprint images** by applying suitable preprocessing techniques such as grayscale conversion, noise removal, and contrast enhancement.
2. **To isolate fingerprint ridge structures** using segmentation and thresholding techniques so that the fingerprint area can be separated from the background.

3. **To improve ridge visibility and continuity** by applying morphological operations and thinning techniques.
4. **To extract basic fingerprint features** such as ridge patterns, ridge endings, bifurcations, or skeletonised ridge structures for analysis.
5. **To evaluate the enhancement result** by comparing the original fingerprint image with the processed image based on clarity, ridge visibility, and feature extraction accuracy.

Literature Review

Study 1 – Fingerprint Image Enhancement Using Gabor Filter

- Applied Adaptive Histogram Equalization to improve fingerprint contrast.
- Used Gabor filtering to reduce noise and enhance ridge and valley patterns.
- Performed binarisation and thinning to produce clearer ridge structures.
- Achieved an average PSNR of approximately **16.87 dB** and **MSE of 87.28**.
- Suitable as a reference because the results are moderate and realistic.
- Limitation: severe noise may still cause broken or inaccurate fingerprint ridges.

Article: Erwin et al. (2019), *The Enhancement of Fingerprint Images Using Gabor Filter*.

<https://iopscience.iop.org/article/10.1088/1742-6596/1196/1/012045/pdf>

Study 2 – Performance Evaluation of Fingerprint Enhancement Algorithms

- Compared Modified Gabor Filter, High Boost Gaussian Filter, and Fuzzy-Based Filter.
- Evaluated the techniques based on noise reduction and fingerprint ridge clarity.
- The Modified Gabor Filter achieved a PSNR of **28.17 dB** and **MSE of 22.27**.
- The results show that the selection of enhancement filters affects fingerprint image quality.
- Suitable as a reference for selecting an appropriate enhancement technique.
- Limitation: enhancement filtering alone does not perform segmentation, thinning, or fingerprint feature extraction.

Article: Mukaila et al. (2019), *Performance Evaluation of Fingerprint Image Enhancement Algorithms*.

<https://fjpas.fuoye.edu.ng/index.php/fjpas/article/view/79/71>

Planning

Member	Preprocessing	Image Processing Technique
Member 1: Image Preprocessing Module	Convert images to grayscale, resize them, normalise intensity, and remove basic noise.	Apply AHE or CLAHE to improve ridge and valley contrast. Compare PSNR and MSE before and after enhancement.
Member 2: Ridge Enhancement Module	Remove noise and adjust brightness before ridge enhancement.	Apply Gabor or Modified Gabor Filter to strengthen ridge patterns. Test parameters to achieve higher PSNR and lower MSE.
Member 3: Ridge Segmentation and Restoration Module	Segment the fingerprint region and remove unnecessary background areas.	Apply thresholding and morphological operations to remove noise and reconnect broken ridges.
Member 4: Feature Extraction and Evaluation Module	Prepare the enhanced image for thinning and feature analysis.	Apply thinning and extract ridge endings and bifurcations. Evaluate using PSNR, MSE, SSIM, and ridge clarity.

**Apply clahe with morphologic apply many as possible
image restoration**

Psnr also can do template matching

**We git the output but the input is still nweed to
clarsify**

<https://www.kaggle.com/datasets/ruizgara/socofing>